LAGUNA STATE POLYTECHNIC UNIVERSITY

Bachelor of Science in Information Technology

Enterprise Operating System Comparison

Windows Server vs. Ubuntu Server vs. Rocky Linux

A practical comparison for choosing a server platform based on workload, skills, support, licensing, and operational needs.

Project Information	Details
Windows Server	Windows Server 2025 Evaluation for the lab comparison
Ubuntu Server	Ubuntu Server 26.04.1 LTS as the current LTS point release listed on 26 October 2026
Rocky Linux	Rocky Linux 10.2 as the current release listed by Rocky Linux on 26 October 2026
Prepared by	[Your Full Name]

Prepared for ITEP 414 - System Administration and Maintenance

Working template: replace bracketed fields and add genuine lab evidence before submission.

1. Comparative Table

Criterion	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Proprietary commercial product. Evaluation edition is available for lab use under Microsoft's evaluation terms; full production use requires appropriate licensing.	Open-source distribution with no OS license fee for standard community use. Optional commercial support and Ubuntu Pro services are available.	Open-source enterprise Linux distribution with no OS license fee. Commercial support may be purchased from third parties.
User interface	Offers Server Core and Desktop Experience options; GUI tools and PowerShell are common administration methods.	Usually deployed with a command-line interface for server workloads; remote administration is common.	Usually deployed with a command-line interface; graphical packages can be installed if needed.
Package management	Windows Update, Server Manager roles/features, MSI and other Windows software deployment methods; PowerShell supports automation.	APT and .deb packages from configured repositories; apt update and apt upgrade are common maintenance commands.	DNF and RPM packages from configured repositories; dnf upgrade is a common maintenance command.
Security	Integrates with Microsoft identity and policy tools; includes Windows security features and can use Secure Boot and Defender technologies depending on configuration.	Security updates, permissions, sudo, firewall tooling, SSH controls, and optional Ubuntu Pro security services; configuration and patching remain essential.	SELinux, firewalld, Linux permissions, SSH controls, and package updates are commonly used; security depends on correct configuration and maintenance.
Performance	Strong fit for Microsoft workloads; resource use varies by installed roles and whether Desktop Experience is used.	Efficient for many web, cloud, automation, and container workloads; actual performance depends on hardware, kernel, workload, and tuning.	Efficient for enterprise Linux workloads that benefit from a stable RHEL-compatible environment; actual performance depends on workload and tuning.
Typical enterprise uses	Active Directory Domain Services, Windows file/print services, IIS, Hyper-V, and Microsoft application workloads.	Web hosting, cloud images, containers, development platforms, automation, and general Linux services.	Enterprise Linux applications, web/database services, infrastructure platforms, and workloads requiring RHEL compatibility.
Advantages	Familiar GUI options, deep Microsoft ecosystem integration, strong administration tools, and vendor support options.	Large community, broad cloud support, familiar APT workflow, frequent documentation, and LTS lifecycle.	RHEL-compatible ecosystem, enterprise-oriented stability, open-source distribution, and a long support lifecycle for major releases.
Disadvantages	Licensing and client-access requirements can add cost; Windows-specific administration and patching processes require skills and planning.	Some commercial applications may require separate vendor certification or support; command-line administration may be unfamiliar to Windows-focused teams.	Smaller ecosystem than some upstream or commercial distributions; RHEL compatibility does not automatically include Red Hat's commercial support contract.

2. Discussion

There is no universally best server operating system. The correct choice depends on application compatibility, staff expertise, support expectations, security requirements, licensing budget, and the operational tools already used by the organization.

Windows Server is often the natural choice when the environment relies on Active Directory, Windows-specific applications, IIS, Hyper-V, or centralized Microsoft management. Its graphical and PowerShell-based administration tools can be advantageous for teams already trained in the Microsoft ecosystem. The trade-off is the need to understand product licensing, edition capabilities, update management, and the resource impact of selected roles and interface options.

Ubuntu Server is a strong general-purpose option for web services, cloud workloads, containers, development environments, and automation. APT provides a straightforward way to update package metadata and install or upgrade software. Ubuntu LTS releases are designed for users who prioritize a longer support window. Organizations should still plan patching, monitoring, backups, identity, firewall rules, and support coverage for the packages they use.

Rocky Linux is designed as an open-source enterprise operating system with bug-for-bug compatibility goals with Red Hat Enterprise Linux. Its DNF/RPM ecosystem and enterprise Linux conventions make it relevant for organizations that require compatibility with software certified for the RHEL family. Administrators should confirm the exact Rocky Linux release, hardware architecture requirements, vendor certification, and lifecycle dates before production deployment.

For the Week 3 learning scenario, Ubuntu Server is appropriate because the assignment specifically requires Ubuntu installation, OpenSSH, and Linux command-line verification. The Windows Server evaluation and Rocky Linux comparison broaden the student's understanding of the differences between a commercial Windows platform and two open-source Linux platforms. This is a learning recommendation, not a benchmark claim; a production decision should be based on workload testing and organizational requirements.

3. Suggested Decision Checklist

- Does the application vendor support the selected OS and release?
- What licensing and support costs apply over the intended service life?
- Does the team have the skills to patch, secure, automate, and troubleshoot the platform?
- What identity, monitoring, backup, logging, and configuration-management tools are already in use?
- What is the supported lifecycle for the exact OS release?
- Has the workload been tested under realistic performance and recovery conditions?

4. References

[1] Microsoft Windows Server 2025 Evaluation Center. https://www.microsoft.com/en-us/evalcenter/download-windows-server-2025. Accessed 26 October 2026.

[2] Microsoft Windows Server release information. https://learn.microsoft.com/en-us/windows/release-health/windows-server-release-info. Accessed 26 October 2026.

[3] Ubuntu current releases. https://ubuntu.com/project/docs/release-team/list-of-releases/. Accessed 26 October 2026.

[4] Ubuntu release cycle and support. https://ubuntu.com/about/release-cycle. Accessed 26 October 2026.

[5] Ubuntu Server package management. https://ubuntu.com/server/docs/how-to/software/package-management/. Accessed 26 October 2026.

[6] Rocky Linux releases. https://docs.rockylinux.org/releases/. Accessed 26 October 2026.

[7] Rocky Linux DNF package manager documentation. https://docs.rockylinux.org/guides/package_management/dnf_package_manager/. Accessed 26 October 2026.

[8] Rocky Linux official site. https://rockylinux.org/. Accessed 26 October 2026.